scientific reports

Check for updates

OPEN

The application of machine learning in the prediction of sports injuries: systematic review and meta-analysis

Sijie Lou[1,6], Houwei Zhu[1,6], Zhanyang He[1,2], Binyong Ye[4,5], Gang Sun[1] & Cuimei Shen[3]✉

Predicting sports injuries is highly challenging, and current machine learning-based models for predicting sports injuries hold promise for addressing the difficulties in sports injury prediction. This study aims to assess the overall performance of existing prediction models in forecasting sports injuries. A comprehensive search was conducted in five databases: PubMed, Web of Science, Scopus, Cochrane Library, and IEEE Xplore. The search covered studies from the inception of each database until December 2025, and the performance of sports injury prediction models was reported. Subgroup and meta-regression analyses were performed based on potential influencing variables. Results: Ten predictive models across ten independent studies were ultimately included in the present analysis. The bivariate random-effects model yielded a pooled sensitivity of 0.79 (95% CI: 0.66–0.87) and a pooled specificity of 0.71 (95% CI: 0.61–0.80) for the included predictive models. The overall diagnostic odds ratio (DOR) was 9.02 (95% CI: 4.31–18.88), demonstrating substantial predictive efficacy. Furthermore, the pooled positive and negative likelihood ratios (PLR and NLR) were 2.72 (95% CI: 1.93–3.83) and 0.30 (95% CI: 0.18–0.49), respectively. Assessment of heterogeneity revealed no significant threshold effect, as evidenced by a weak correlation between logit sensitivity and logit specificity (r = -0.11) and a non-significant asymmetry parameter in the HSROC model (beta= -0.30, P = 0.372). Potential factors such as sample size, model type, validation method, and injury type may influence model performance. Conclusion: Existing machine learning-based sports injury prediction models exhibit considerable methodological differences, and these methodological barriers affect the improvement and enhancement of model performance. It is recommended to apply imbalance handling to datasets with small sample sizes, use k-fold cross-validation to prevent overly optimistic prediction performance, and combine time-series data with other data for feature extraction. These suggestions need to be further validated through larger-scale studies to improve the performance of sports injury prediction models.

Keywords Sports injury prediction, Machine learning, Meta-analysis

Sports injuries refer to injuries that occur during physical activities, including muscle injuries, bone injuries, ligament injuries, etc[1,2].Sports injuries are common among athletes of all levels[3]. The occurrence of sports injuries is related to various factors such as practice duration, lack of warm-up activities, improper training load, and lack of technical guidance[4], making the causes highly complex. Sports injuries not only impact athletes' physical and mental health and performance, but also increase the social healthcare burden and reduce athletes' quality of life[4–6]. Consequently, accurately assessing injury risk in athletes to implement tailored prevention strategies and reduce injury incidence remains a persistent frontier in sports medicine[7,8].

In existing studies, sports injury prediction models are primarily established using two approaches: traditional statistics and machine learning[9,10]. Although many studies have established sports injury prediction models using statistical methods, since sports injuries result from multiple interacting factors, prediction models built using traditional statistical methods struggle to capture the impact of these non-linear relationships on sports injury

[1]College of Physical Education and Health Sciences, Zhejiang Normal University, Jinhua, China. [2]School of Sport Science, Beijing Sport University, Beijing, China. [3]Department of Business Administration, Shanghai University of Finance and Economics Zhejiang College, Jinhua, China. [4]Faculty of Health Sciences and Sports, Macao Polytechnic University, Macao, China. [5]Faculty of Sports Science, Ningbo University, Ningbo, China. [6]Sijie Lou and Houwei Zhu contributed equally to this work. ✉email: shencuimei@shufe-zj.edu.cn

risk, making it difficult to achieve optimal prediction performance[11–14]. Given its capacity to process extensive datasets and model complex non-linear interactions among physiological and biomechanical variables, machine learning is widely recognized as a highly suitable approach for sports injury prediction. Currently, there are many studies on machine learning-based sports injury prediction models. Maria et al. established a prediction model for lower limb musculoskeletal injuries in athletes based on machine learning[15], and Juri et al. built a sports injury prediction model using machine learning algorithms like SVM, DT, and KNN to predict anterior cruciate ligament injuries in athletes[16]. However, the performance of the prediction models varies significantly. Additionally, a previous review also pointed out that sports injury prediction models built using machine learning algorithms do not show significant advantages in prediction performance[17].Another recent review on machine learning-based sports injury prediction models also pointed out that high methodological heterogeneity may be one of the reasons for the inconsistent performance of existing sports injury prediction models[18]. Recent studies emphasize the necessity of standardizing AI evaluation protocols to establish a methodological baseline for the rigorous appraisal of emerging technologies, thereby ensuring the evidence-based implementation of artificial intelligence in athletic training[19]. However, due to the absence of a quantitative synthesis evaluating the performance metrics of existing machine learning-based models, how specific methodological factors influence predictive efficacy remains poorly understood.

In summary, this study conducts an extensive search of empirical research literature on the use of machine learning to establish sports injury prediction models, systematically organizes and synthesizes the findings, and provides a comprehensive evaluation of the current empirical research results, aiming to gain a deeper understanding of the effectiveness and practical application value of these sports injury prediction models.

Method

This systematic review was prospectively registered in the PROSPERO database (Registration No. CRD42024573751) and conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses for Diagnostic Test Accuracy (PRISMA-DTA) statement[20].

Search strategy

Based on the PRISMA statement, a comprehensive search was conducted in five databases: PubMed, Web of Science, Scopus, Cochrane Library, and IEEE Xplore. The search period ranged from the inception of each database until December 2025, final literature search was conducted on December 15, 2025. The main search keywords were: (1) athlete OR athletes OR individual sports OR individual sport (2) injury prediction OR injury forecasting OR injury detection OR injury risk OR injury prevention (3) machine learning OR learning machine OR supervised learning OR semi-supervised learning OR unsupervised learning OR neural networks OR deep learning OR AI OR artificial intelligence OR neural network OR CNN OR ANN OR digital phenotyping OR training load OR workload OR internal load OR external load OR ACWR OR kinematic OR kinetic OR biomechanic OR GPS OR Global Positioning System OR wearable OR IMU OR inertial measurement unit OR surface electromyography .

Inclusion Criteria: To be included in this review, studies had to meet the following criteria: (1) empirical research enrolling athletes as the target population; (2) aiming to predict the occurrence of sports-related injuries; (3) reporting core predictive performance metrics, including accuracy, area under the curve (AUROC), sensitivity, and specificity; and (4) published as original articles in peer-reviewed, English-language journals. Exclusion Criteria: The exclusion criteria were as follows: (1) predictive models constructed using non-machine learning algorithms; and (2) conference abstracts or articles for which the full text was inaccessible.

Manual searches were conducted for the references of the included studies and systematic reviews to avoid missing relevant research. The studies were uploaded to EndNote, and duplicate studies were removed. Two authors independently screened the titles and abstracts of all studies and included those that met the inclusion criteria. The studies passing the initial screening were independently reviewed in full by two authors. If disagreements arose, the two authors resolved them through discussion; otherwise, a third independent reviewer addressed the conflicts. Our protocol has been registered on PROSPERO (CRD42024573751) and follows the PRISMA guidelines.

Data extraction and management

This study was conducted and reported in adherence to the TRIPOD-AI guidelines, Two researchers.

(LSJ and HZY) independently extracted all data using Excel 2010, first performing literature screening according to the PRISMA statement[21]. (1) Removal of duplicate literature: All retrieved references were imported into EndNote 20 software, and duplicates were removed. (2) Initial screening: Two researchers removed irrelevant studies based on the title and abstract. (3) Second screening: Two researchers reviewed the full text of the studies and excluded those that did not meet the criteria. (4) Inclusion: The relevant studies obtained from databases and other sources were summarized, and studies that met the criteria were included. Data extraction was then performed on the included studies, extracting the following data: article title or DOI, first author, publication year, study subjects and basic information, data source, sample size, study design, feature extraction, dataset processing, and outcome performance indicators related to the prediction model (accuracy, AUROC, true positive rate, true negative rate, false negative rate, and false positive rate, This enabled the derivation of the true positive (TP), false positive (FP), true negative (TN), and false negative (FN) values.

Risk of bias and quality assessment

Reporting quality was appraised in accordance with the TRIPOD-AI guidelines, whereas risk of bias was evaluated using the PROBAST-AI tool across four principal domains: participants, predictors, outcome, and analysis[22]. Three investigators (R.K., V.S., and O.R.) independently conducted all assessments; any discrepancies

were resolved through iterative discussion to ensure inter-rater consistency. The TRIPOD-AI checklist, comprising approximately 25 to 28 items, was utilized to evaluate reporting completeness, explicitly covering aspects such as data preprocessing, feature selection, hyperparameter tuning, and model interpretability. Based on a predefined scoring rubric, the included studies were stratified into high (≥ 80%), moderate (60%–79%), or low (< 60%) quality. Furthermore, employing 25 to 30 signaling questions, the PROBAST-AI tool identified potential sources of bias—such as participant selection, class imbalance, and the absence of external validation—ultimately grading the overall risk as low, moderate, or high. Finally, the certainty of evidence was assessed utilizing an adapted GRADE framework. Standard domains (risk of bias, consistency, directness, and precision) were applied to the model-derived outcomes to reflect the reliability and clinical interpretability of the predictive performance, with the final certainty rated as high, moderate, low.

Data analysis

To evaluate the performance of machine learning algorithms in predicting sports injuries, the AUROC, sensitivity, specificity, and their 95% confidence intervals (CI) from the included studies were used to assess the overall performance of the prediction models. For the moderator analysis and meta-regression involving sample size, the median sample size across the included studies served as the threshold for stratification. For investigations that failed to report a complete confusion matrix, the relevant parameters were indirectly derived utilizing a binormal model, based on the reported area under the curve (AUROC) alongside the exact counts of injured and healthy cases. Recognizing that the majority of the included studies evaluated multiple machine learning algorithms on identical datasets, extracting all models would violate the statistical assumption of sample independence. To prevent disproportionately amplifying the weight of any single study cohort while concurrently mitigating the optimistic bias associated with extracting the 'best-performing model', we implemented an objective, a priori extraction criterion. Specifically, we extracted the data corresponding exclusively to the primary model explicitly highlighted in the abstract or main conclusion of each included study. This methodological precaution ensures that the aggregated metrics reflect the algorithms deemed most robust and generalizable by the original investigators, rather than theoretical peak performances driven by overfitting. This methodological precaution was implemented to prevent violating the statistical assumption of sample independence and to avoid disproportionately amplifying the weight of any single study. Additionally, we qualitatively appraised specific methodological indicators within the included literature, such as techniques for addressing class imbalance, the application of model interpretability analyses, and cohort characteristics. We performed the meta-analysis utilizing the Hierarchical Summary Receiver Operating Characteristic (HSROC) model, a hierarchical regression framework that extends the bivariate model to better account for within-study (Level 1) and between-study (Level 2) variability in true positive and false positive rates[23]. This approach yielded pooled estimates for sensitivity and specificity, along with their corresponding 95% confidence intervals (CIs) and prediction intervals (PIs).

Results

Study selection process

A total of 1,548 records were initially identified from the electronic databases using the predefined search strategy. After importing the citations into EndNote software, 399 duplicate records were removed. Following a preliminary screening of titles, abstracts, and keywords, an additional 818 articles were excluded. Of the remaining 331 articles, a comprehensive full-text assessment led to the exclusion of 318 articles, leaving 12 studies eligible for inclusion in the systematic review. Among these 12 studies, 10 provided extractable effect sizes, 95% confidence intervals (CIs), or calculable baseline data. Consequently, these 10 studies were incorporated into the quantitative meta-analysis. The primary reason for excluding the remaining two articles from the meta-analysis was the reporting of exclusively qualitative metrics. The detailed literature search and selection process is illustrated in Fig. 1.

Basic characteristics of the included studies

A total of 12 articles[8,24–33] were identified for this systematic review, with 10 models from 10 studies meeting the criteria for meta-analysis. These investigations, published between 2020 and 2025, represented a cumulative sample size of 12,249 across the United States ($n = 5$), Europe ($n = 4$), and Australia ($n = 1$). Regarding injury classification, one study focused on brain injury and one on anterior cruciate ligament (ACL) injury, while the remainder predicted non-specific sports injuries. Random Forest was the most widely utilized algorithm ($n = 5$), followed by XGBoost ($n = 4$) and Support Vector Machine ($n = 1$). Notably, external validation was absent in all included studies. Internal validation through k-fold cross-validation was explicitly reported in six models, whereas the protocols for the remaining studies were not disclosed. Additionally, model interpretability was addressed in only four studies. The detailed characteristics of the included studies are summarized in Table 1.

Overall performance

The bivariate random-effects model yielded a pooled sensitivity of 0.79 (95% CI: 0.66–0.87) and a pooled specificity of 0.71 (95% CI: 0.61–0.80) for the included predictive models. The overall diagnostic odds ratio (DOR) was 9.02 (95% CI: 4.31–18.88), demonstrating substantial predictive efficacy. Furthermore, the pooled positive and negative likelihood ratios (PLR and NLR) were 2.72 (95% CI: 1.93–3.83) and 0.30 (95% CI: 0.18–0.49), respectively. Assessment of heterogeneity revealed no significant threshold effect, as evidenced by a weak correlation between logit sensitivity and logit specificity ($r = -0.11$) and a non-significant asymmetry parameter in the HSROC model (beta= -0.30, $P = 0.372$). Although no significant threshold effect was detected, the model revealed substantial non-threshold heterogeneity among the included studies. The variance of logit sensitivity (0.96) was notably higher than that of logit specificity (0.53), indicating considerable inconsistency in the models'

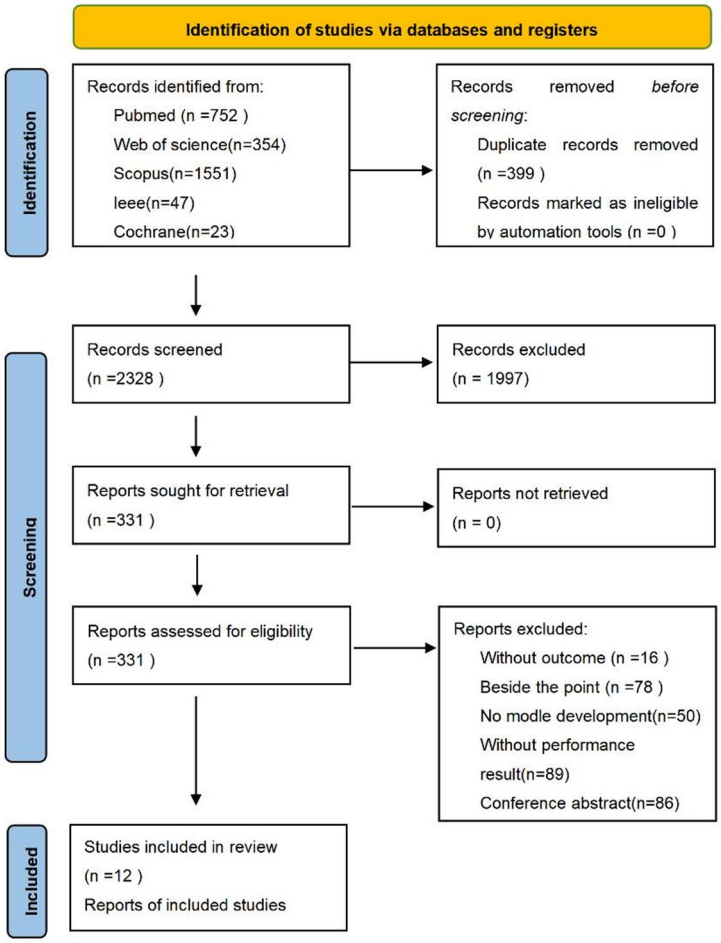


Fig. 1. PRISMA flow diagram of study selection process.

First author	Target condition	Sample size	Algorithm	Study design	Type of internal validation	Performance metrics
Sofie et al. (2020)	All sport injury	3044	XGBoost	Retrospective	K-fold cross-validation	AUC = 0.678 (0.004) Spec = 0.746 Sens = 0.504 (0.05)
Simon et al. (2022)	All sport injury	355	RF	NA	Holdout method	NA
Susanne et al. (2022)	ACL injury	286	RF	Prospective	K-fold cross-validation	AUC = 0.63 (0.51, 0.69)
Yuanqi Huang et al. (2022)	LE injury	1811	XGBoost	NA	10-fold cross-validation	AUC = 0.62 (0.01)
Yining Lu et al. (2022)	LEMS	2103	RF, XGBoost	Case-control	10-fold cross-validation	AUC (RF)=0.906 (0.899, 0.911);AUC (XGB)=0.905 (0.896, 0.92)
Yi Chung et al. (2022)	All sport injury	NA	SVM	Retrospective	Holdout method	NA
Francisco et al. (2023)	LE-ST	260	RF	Prospective	Holdout method	AUC = O.700 (0.088) TP = 53.7% TN = 73.9% F-score = 0.38 (0.015)
Angus Shaw et al. (2023)	MTSS	222	RF	Prospective	Holdout method	AUC = 0.98; CA = 0.96
Bryan et al. (2020)	All sport injury	2322	XGBoost	Prospective	10-fold cross-validation	AUC (XGB)=0.948;AUC (LR)=0.937
Joel et al. (2020)	Concussion	595	XGBoost	Prospective	NA	AUC = 0.73 (0.70, 0.76)
Mauro et al. (2020)	All sport injury	1118	SVM	Retrospective	4-fold cross-validation	AUC = 0.84 (0.05)
Oliver et al. (2020)	All sport injury	355	RF	Prospective	NA	AUC = 0.663;Spec = 0.742 (0.03)Sens = 0.556 (0.01)

Table 1. Model development methodology of included studies ($n = 10$).

capacity to correctly identify injured athletes. Furthermore, the variance of the overall accuracy parameter (1.27) corroborated a high degree of heterogeneity in predictive performance. Consequently, moderator analyses and meta-regression were performed to elucidate the specific methodological sources driving this variance.

Subgroup analysis

When conducting the subgroup analysis based on sample size, this study used the median sample size of all included studies as the criterion to divide the studies into a large sample size group and a small sample size group (Fig. 2b, c). For the large sample size group, the pooled sensitivity was 0.85 (95% CI: 0.74–0.92), the pooled specificity was 0.70 (95% CI: 0.50–0.85), the diagnostic odds ratio (DOR) was 13.30 (95% CI: 6.24–28.33), the positive likelihood ratio (LR+) was 2.85 (95% CI: 1.64–4.96), the negative likelihood ratio (LR-) was 0.21 (95% CI: 0.13–0.35), and the variance of diagnostic accuracy was 0.64. In the small sample size group, the pooled sensitivity was 0.70 (95% CI: 0.49–0.85), the pooled specificity was 0.72 (95% CI: 0.63–0.79), the DOR was 6.02 (95% CI: 1.84–19.69), the positive likelihood ratio (LR+) was 2.48 (95% CI: 1.52–4.05), the negative likelihood ratio (LR-) was 0.41 (95% CI: 0.20–0.82), and the variance of diagnostic accuracy was 1.36.

When conducting the subgroup analysis based on the validation method, this study divided the included studies into a k-fold cross-validation group and an unused cross-validation group (Fig. 2d, e). For the k-fold

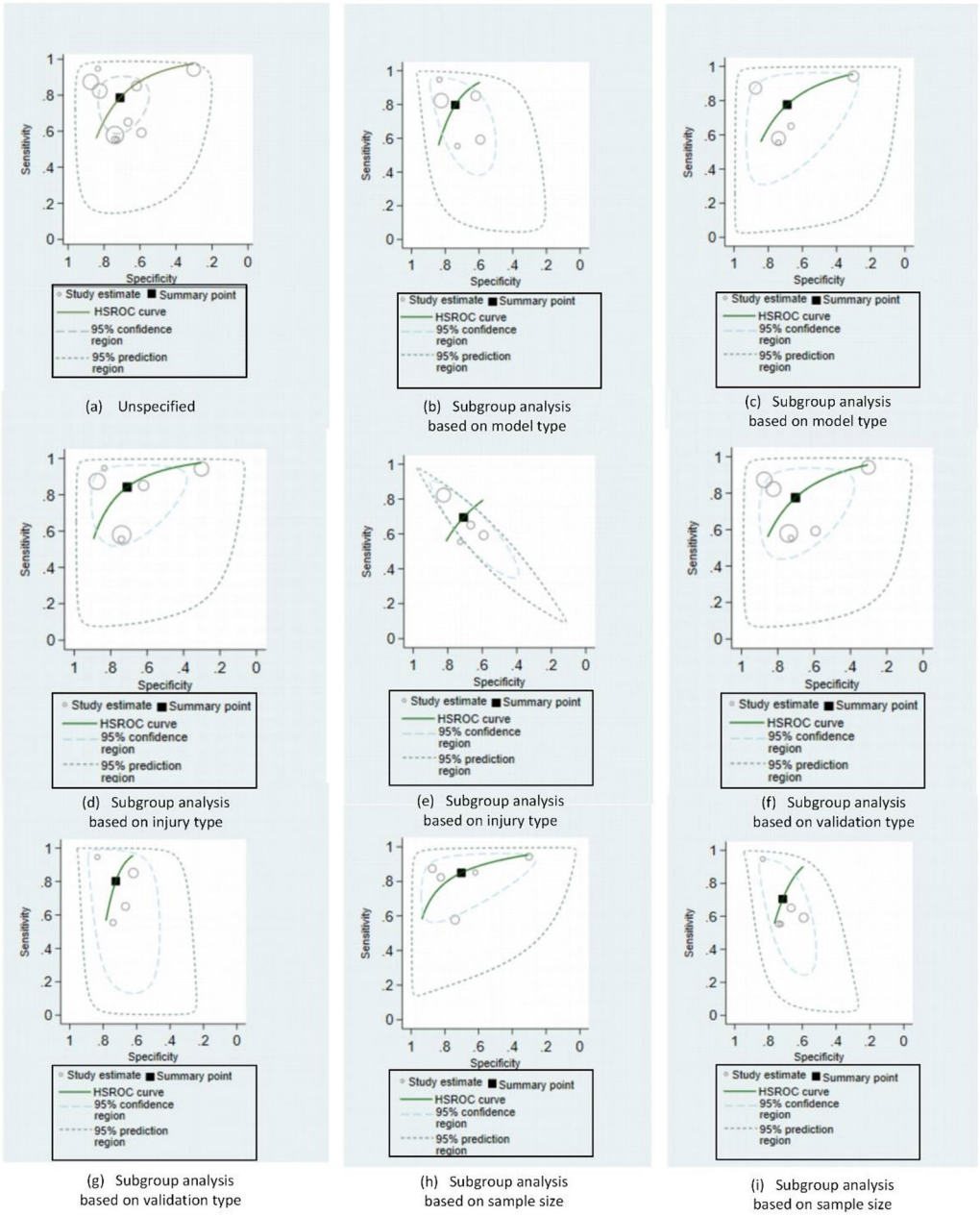


Fig. 2. Hierarchical ROC curves.

cross-validation group ($n=6$), the pooled sensitivity was 0.78 (95% CI: 0.61–0.88), the pooled specificity was 0.70 (95% CI: 0.54–0.83), the diagnostic odds ratio was 8.15 (95% CI: 3.33–19.94), the positive likelihood ratio was 2.60 (95% CI: 1.61–4.21), the negative likelihood ratio was 0.32 (95% CI: 0.18–0.57), and the variance of diagnostic accuracy was 1.20. In the unused cross-validation group ($n=4$), the pooled sensitivity was 0.80 (95% CI: 0.59–0.92), the pooled specificity was 0.72 (95% CI: 0.65–0.79), the DOR was 10.63 (95% CI: 3.07–36.84), the positive likelihood ratio was 2.91 (95% CI: 1.96–4.31), the negative likelihood ratio was 0.27 (95% CI: 0.11–0.66), and the variance of diagnostic accuracy was 0.93.

When conducting the subgroup analysis based on injury type, this study divided the included studies into a whole-body injury group and a lower extremity injury group (Fig. 2f, g). For the whole-body injury group ($n=6$), the pooled sensitivity was 0.84 (95% CI: 0.70–0.93), the pooled specificity was 0.71 (95% CI: 0.54–0.83), the diagnostic odds ratio was 13.04 (95% CI: 4.95–34.37), the positive likelihood ratio was 2.90 (95% CI: 1.76–4.77), the negative likelihood ratio was 0.22 (95% CI: 0.11–0.44), and the variance of diagnostic accuracy was 1.36. In the lower extremity injury group ($n=4$), the pooled sensitivity was 0.69 (95% CI: 0.59–0.78), the pooled specificity was 0.71 (95% CI: 0.61–0.79), the DOR was 5.51 (95% CI: 2.31–13.16), the positive likelihood ratio was 2.38 (95% CI: 1.54–3.67), the negative likelihood ratio was 0.43 (95% CI: 0.28–0.67), and the variance of diagnostic accuracy was 0.75.

When conducting the subgroup analysis based on model type, this study divided the included studies into a Random Forest group and a non-Random Forest (non-RF) group (Fig. 2h, i). For the RF group ($n=5$), the pooled sensitivity was 0.80 (95% CI: 0.63–0.90), the pooled specificity was 0.74 (95% CI: 0.64–0.82), the diagnostic odds ratio was 11.21 (95% CI: 3.46–36.33), the positive likelihood ratio was 3.07 (95% CI: 1.92–4.90), the negative likelihood ratio was 0.27 (95% CI: 0.13–0.58), and the variance of diagnostic accuracy was 1.52. In the non-RF group ($n=5$), the pooled sensitivity was 0.78 (95% CI: 0.58–0.90), the pooled specificity was 0.69 (95% CI: 0.50–0.83), the DOR was 7.69 (95% CI: 3.12–18.95), the positive likelihood ratio was 2.49 (95% CI: 1.53–4.07), the negative likelihood ratio was 0.32 (95% CI: 0.17–0.62), and the variance of diagnostic accuracy was 0.96.

Literature Quality, Bias Risk Assessment, and Sensitivity Analysis.

Based on the PROBAST-AI tool, 7 out of the 10 included studies were assessed as having a low risk of bias, while the remaining 3 studies showed an unclear risk (Fig. 3). The unclear risk of bias was primarily attributed to the lack of measures to control overfitting and unclear descriptions of the datasets. Publication bias among the included studies was evaluated using Deeks' funnel plot asymmetry test, which is specifically recommended for diagnostic test accuracy meta-analyses. Visual inspection of the funnel plot demonstrated a relatively symmetrical distribution of the data points around the regression line. Furthermore, the quantitative analysis yielded a p-value of 0.104, which indicates that there was no significant publication bias present in this meta-analysis. ($P>0.05$) (Fig. 4).

Discussion

Similar to this study, Garrett et al. conducted a systematic review of musculoskeletal injury prediction models in sports and identified methodological flaws and significant bias risks in the studies they included. The study pointed out that the lack of prior sample size calculations, the opacity of model codes, and small sample sizes might be the main factors contributing to these issues[17]. However, this study did not assess the performance of the prediction models and lacked a quantitative analysis of the model outcome indicators, thus lacking sufficient evidence to show the extent to which these methodological flaws affect the final performance of sports injury prediction models. In another broad review[18], researchers assessed the performance of the models by comparing

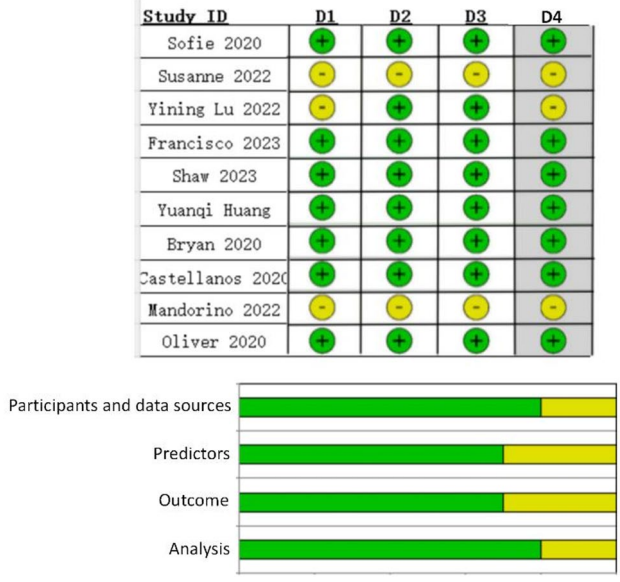


Fig. 3. Summary of the overall risk of bias.

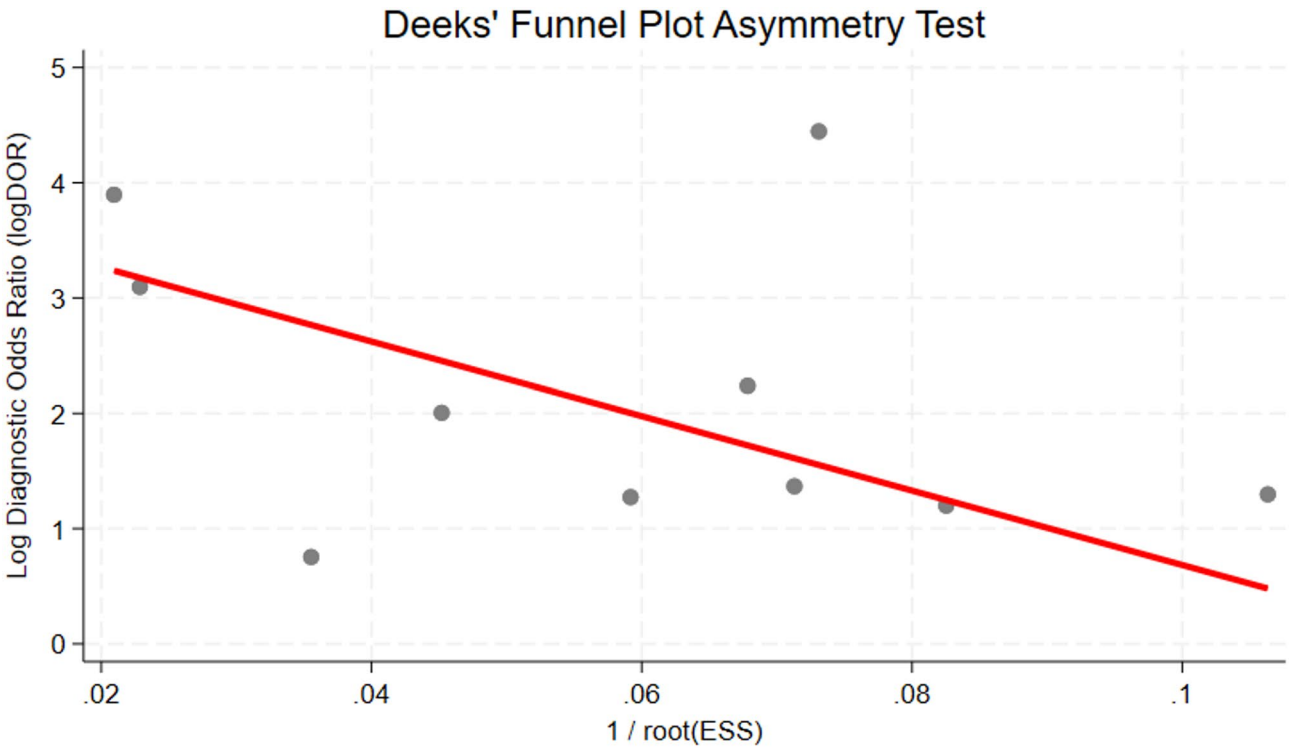


Fig. 4. Deeks' funnel plot.

the AUROC values from two studies. However, this method still has significant limitations. First and foremost, such comparisons are highly susceptible to spectrum bias and population heterogeneity. The discriminative ability of a model fundamentally depends on the diagnostic difficulty of the specific cohort; consequently, a model evaluated on a case-control population with distinct characteristics will naturally yield an inflated AUC compared to a model tested on a real-world cohort with ambiguous clinical presentations. Secondly, disparities in validation strategies can also significantly confound these metrics. Methodologically, it is invalid to compare an AUC derived from internal validation which is prone to overfitting with an AUC obtained from an independent, spatio-temporal external cohort. Third, isolated numerical comparisons preclude the possibility of conducting rigorous statistical testing. Therefore further quantitative analysis of existing studies is needed.

This systematic review and meta-analysis demonstrated the performance of machine learning algorithms in predicting sports injuries. In the overall meta-analysis of all included studies, the machine learning algorithms demonstrated a pooled sensitivity of 0.79 (95% CI: 0.66–0.87) and a pooled specificity of 0.71 (95% CI: 0.61–0.80) for predicting sports injuries. The overall diagnostic odds ratio was 9.02 (95% CI: 4.31–18.88). Furthermore, the pooled positive likelihood ratio was 2.72 (95% CI: 1.93–3.83), and the pooled negative likelihood ratio was 0.30 (95% CI: 0.18–0.49), indicating a moderate to good overall predictive value. According to the quality assessment, a minority of the included studies presented an unclear risk of bias.

Regarding heterogeneity, the variance of overall diagnostic accuracy was 1.27, reflecting substantial between-study heterogeneity. However, the asymmetry parameter (beta) from the HSROC model was not statistically significant (beta = -0.30, $P = 0.372$), suggesting that the observed heterogeneity was not primarily driven by a threshold effect. Given this substantial non-threshold heterogeneity, further subgroup analyses and assessment of modifying variables were necessary to identify the potential sources of variation across the included studies.

Subgroup analysis

Our subgroup analysis based on sample size revealed critical insights into the training efficacy and stability of machine learning (ML) algorithms for predicting sports injuries. Notably, models trained on larger cohorts (≥ median sample size) demonstrated a superior pooled sensitivity (0.85 vs. 0.70) and diagnostic odds ratio (DOR: 13.30 vs. 6.02) compared to those trained on smaller cohorts, while maintaining a comparable specificity (0.70 vs. 0.72). This study further conducted a meta-regression on sample size, but the results did not find an effect of sample size on prediction model performance. Empirical observations from the subgroup analysis revealed a numerical trend where larger sample cohorts yielded superior pooled sensitivity (0.85 vs. 0.70). However, limited by the insufficient statistical power typical of meta-regressions with a small number of included studies, we failed to detect a statistically significant association between sample size and predictive performance ($p > 0.05$). This is similar to the findings of Sun et al.[34], who noted that there is no significant correlation between sample size and model performance. In the absence of guidance for selecting an appropriate sample size, the research by Riley et al.[35] can provide some reference. This study also found that all included studies lacked information on sample size calculations, and therefore, these prediction models may have undetected overfitting risks[17], which

could reduce the generalizability and usefulness of the models[36]. Additionally, this study found that the articles had datasets from a single source, typically from one club or league. This flaw in the sample source can also introduce a certain bias risk in the studies. In addition to conducting a meta-analysis of quantifiable indicators, this study also conducted a comprehensive evaluation of the methodological indicators of the included studies and found methodological differences in the process of building prediction models. First, this study found that the included literature generally had an issue of sample size imbalance, with sports injury events being fewer than healthy events. However, machine learning prediction models must be trained on balanced datasets[30]. Only 6 of the included studies reported handling dataset imbalance. When handling sample size imbalance, most studies used oversampling or undersampling techniques. While these methods can improve the imbalance in the dataset, their impact on model performance is difficult to measure. Furthermore, since most studies have fewer injury instances when building their datasets, smaller sample sizes have always been a potential issue limiting the prediction performance of sports injury models[37]. In predicting sports injuries—where positive cases (injuries) often constitute the minority class—sufficient sample sizes are imperative for algorithms to accurately capture the complex, non-linear predictive patterns without being misled by noise. Smaller datasets often fail to provide enough injury events for the model to learn generalizable features, leading to poor sensitivity and a higher false-negative rate. Furthermore, an in-depth evaluation of the HSROC model parameters elucidated the sources of instability in small-sample studies. The variance of overall diagnostic accuracy (s2alpha) in the small-sample subgroup (1.369) was more than twice that of the large-sample subgroup (0.644), accompanied by a remarkably wide 95% confidence interval for sensitivity (0.49–0.85). Intriguingly, the asymmetry parameter (beta) indicated a statistically significant threshold effect exclusively in the small-sample subgroup ($P=0.036$), whereas no such effect was observed in larger studies ($P=0.522$). This suggests that researchers utilizing small cohorts may have explicitly or implicitly manipulated decision thresholds (cut-off values) to optimize apparent performance, inadvertently driving up between-study heterogeneity and exacerbating the risk of overfitting.

Consequently, our findings strongly advocate that future studies aiming to develop ML-based sports injury prediction models must ensure adequate sample sizes. Models derived from small cohorts should be interpreted with extreme caution, as their predictive performance is likely unstable and highly threshold-dependent.

Our subgroup analysis stratified by validation methodology provided critical insights into the methodological robustness of the included studies. Models that did not employ cross-validation exhibited ostensibly superior predictive performance, with a slightly higher pooled sensitivity (0.80 vs. 0.78) and a noticeably higher diagnostic odds ratio (10.63 vs. 8.15) compared to those utilizing k-fold cross-validation. However, from a machine learning perspective, this apparent superiority must be interpreted with skepticism, as it is highly indicative of optimism bias and over fitting .This is consistent with the findings of Garrett et al.[17], who noted that prediction models often show more optimistic results in the test set during the study[38], and that k-fold cross-validation helps control this optimistic bias[39]. The internal validation method of k-fold cross-validation is widely used in prediction model studies to reduce the risk of overfitting in small sample prediction models[40]. Additionally, external validation is a major flaw in research on machine learning prediction models for sports injuries. None of the studies included in this research performed external validation. External validation largely defines the effectiveness of prediction models in independent populations with different characteristics. Studies that did not implement external validation are severely limited in terms of generalizability and reproducibility[41,42]. External validation is crucial not only for the usefulness of prediction models[43], but also for reducing research waste[44]. Therefore, the real-world applicability of existing prediction models remains unclear, and research methods in this area urgently need improvement[17]. In machine learning, algorithms are intrinsically prone to memorizing the training dataset. Without rigorous internal validation mechanisms, such as k-fold cross-validation, the reported diagnostic accuracy merely reflects the model's performance on the data it has already "seen," leading to artificially inflated metrics that frequently fail to generalize to unseen cohorts. Conversely, k-fold cross-validation iteratively evaluates the model on independent data folds, providing a more conservative, realistic, and robust estimation of the model's true clinical utility. Further more, the subgroup lacking cross-validation displayed a wider 95% confidence interval for sensitivity (0.59–0.92) and a borderline significant threshold effect (beta = -1.267, $P=0.065$) despite the small number of studies. This suggests considerable instability and implies that researchers might have manipulated decision thresholds to maximize apparent performance in the absence of a validation set. Therefore, we strongly emphasize that future prognostic models for sports injuries must routinely incorporate stringent internal validation frameworks—or ideally, independent external validation—to ensure reproducibility and safe clinical translation.

Our subgroup analysis stratified by injury type yielded insightful findings regarding the target specificity of machine learning models. Models predicting general or whole-body injuries (without specific anatomical distinctions) exhibited a substantially higher pooled sensitivity (0.84 vs. 0.69) and overall diagnostic odds ratio (DOR: 13.04 vs. 5.51) compared to those specifically targeting lower extremity injuries, while maintaining an identical specificity (0.71). Related research has pointed out that different types of sports injuries have significant differences in mechanisms, which can lead to the model's inability to identify risk factors, thereby reducing its predictive performance.

Clinically and biomechanically, this discrepancy is highly interpretable. General sports injuries are frequently driven by systemic risk factors, such as acute spikes in training load, overall fatigue, and sleep deprivation, which are relatively easy for machine learning algorithms to capture from global monitoring data. Conversely, predicting a specific regional injury—such as lower extremity trauma—requires highly granular, localized biomechanical and neuromuscular data (e.g., joint kinematics, specific muscle imbalances). If the included studies primarily relied on macroscopic workload features rather than localized biomechanical markers, the algorithms would naturally struggle to accurately identify the specific risk of lower extremity injuries, leading to the observed drop in sensitivity. Sofie et al.[24] selected features through indicators in training logs, but the correlation between these indicators and sports injuries has not been verified. The same issue was also present

in the study by Lu et al.[26]. The feature extraction work by Francisco et al.[25] can provide useful reference. This study first identified neuromuscular status, psychological factors, and sleep quality as potential risk factors for sports injuries, and by collecting kinematic and dynamic data during jumps and psychological scales, selected multidimensional features to avoid missing key features. Additionally, Sofie et al. found that combining athletes' time-series data with motion frequency clustering data led to higher model prediction performance. Time-series features of sports injuries may be key to explaining injury occurrences, but most current studies lack research on time-series features[1,24,45]. A critical methodological limitation across the synthesized literature is the pervasive reliance on static, cross-sectional datasets, which fundamentally restricts the temporal validity of the evaluated machine learning algorithms. Sports injuries are rarely the consequence of a single baseline deficit; rather, they manifest from the non-linear, dynamic accumulation of fatigue, acute-on-chronic workload imbalances, and progressive biomechanical deterioration. By predominantly utilizing discrete snapshots of data—such as pre-season anthropometrics or isolated historical injury records—current models fail to capture these continuous hazard trajectories. Consequently, while these static algorithms may successfully identify general baseline susceptibility, they lack the temporal acuity required to forecast the precise onset of structural failure (i.e., when the injury will occur). To transition from theoretical risk profiling to actionable, real-time injury prevention, future predictive modeling must prioritize dynamic training load monitoring and continuous time-series features. This paradigm shift will require the implementation of advanced frameworks, such as recurrent neural architectures or functional data analysis, capable of mapping the complex, time-varying biomechanical signals that immediately precede an injury event. However, simply concatenating time-series metrics with static baseline features is insufficient; this approach requires a robust theoretical foundation to prevent algorithms from memorizing temporal artifacts instead of authentic physiological rhythms. As elegantly articulated in the multivariate framework proposed by Dhahbi and Dergaa, incorporating explicit "temporal control designs" within the machine learning architecture is imperative for biological signal processing[46]. By implementing temporally aware feature extraction—such as functional data analysis for continuous kinetic signals—and chronological cross-validation, these control designs mathematically force the algorithm to evaluate rhythmic specificity. This enables the model to accurately differentiate between random background noise and genuine, cumulative physiological degradation. Consequently, establishing these temporal controls fundamentally guarantees that the algorithms map true biological hazard trajectories, thereby progressively enhancing longitudinal predictive accuracy and safeguarding against overfitting. While restricted sample sources may introduce certain bias risks, it is crucial to recognize that targeted injury profiling within professional, homogeneous cohorts serves as a fundamental prerequisite for constructing robust machine learning training sets. For instance, the recent retrospective cohort study by Dhahbi et al. precisely delineated the epidemiological injury profile of male police cadets during their initial training phase[47]. By meticulously documenting injury incidence, anatomical sites, and mechanisms under standardized physical workloads, such studies effectively filter out unmeasured confounding variables prevalent in highly heterogeneous populations. Consequently, this rigorous epidemiological profiling provides the high-fidelity, context-specific baseline data necessary for predictive algorithms to accurately map non-linear decision boundaries and learn authentic injury patterns before generalizability can be achieved across broader domains. Analyzing the variance of overall diagnostic accuracy (s2alpha) revealed that the between-study heterogeneity was substantially higher in the whole-body injury subgroup (s2alpha = 1.360) than in the lower extremity subgroup (s2alpha = 0.746). This methodologically validates that "mixed injury types" inherently introduce significant clinical heterogeneity. Restricting the prediction target to the lower extremities effectively homogenized the clinical context, thereby reducing inter-study variance, albeit at the cost of reduced overall predictive accuracy. Therefore, future studies aiming to predict specific regional injuries must ensure that their feature engineering incorporates localized, site-specific risk factors rather than relying solely on generalized load metrics.

Limitations

One limitation of this study is the small number of included studies, with only 10 articles considered. Secondly, although this study conducted subgroup analysis on some potential methodological factors, there are still other methodological differences in the included studies that cannot be quantified but may negatively affect prediction performance. The number of studies on sports injury prediction models will continue to increase, and these differences may reach a consensus as the number of studies grows. A notable methodological limitation of this meta-analysis is the reliance on a median split for sample size stratification, rather than the Events Per Variable (EPV) ratio. EPV is the recognized standard for assessing the overfitting risk of multivariable algorithms. However, pervasive reporting deficits in the primary sports biomechanics literature—specifically the failure to disclose the initial number of candidate predictors prior to dimensionality reduction—rendered precise EPV calculations mathematically unfeasible. Consequently, total sample size was utilized as the most reproducible proxy. Future studies must rigorously adhere to TRIPOD-AI reporting guidelines, explicitly detailing both candidate feature dimensions and absolute minority class events, to facilitate robust methodological evaluation. However, currently, the methodological differences across these studies remain significant.

Conclusion

In summary, existing machine learning-based sports injury prediction models have significant methodological differences, and these methodological barriers can affect the improvement and enhancement of model performance. Beyond the fundamental necessity for larger-scale empirical validation, it is imperative to recognize that artificial intelligence is currently initiating a profound paradigm shift in both regional and global sports injury prediction strategies. Particularly within high-contact team sports, advanced machine learning frameworks are transcending localized, traditional workload monitoring to map complex, multidimensional

risk profiles across diverse athletic populations. As recently elucidated by Gaddour et al., who examined how AI is shaping present and future prediction models on a trans-regional scale[6], these intelligent systems are fundamentally transforming risk assessment from static estimations into dynamic, globally adaptable prevention protocols. Consequently, future research must not only strive to expand dataset dimensions but also explicitly embrace this globalized AI perspective, leveraging cross-regional data integration to develop truly robust, scalable, and universally applicable injury forecasting frameworks. It is recommended to handle dataset imbalance when the sample size is small, use k-fold cross-validation to prevent overly optimistic prediction performance, and combine time-series data with other data for feature extraction. These suggestions need to be further validated through larger-scale studies to improve the performance of sports injury prediction models.

Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files. Summary data extracted from included studies are provided in the supplementary tables.

Received: 6 May 2026; Accepted: 22 June 2026
Published online: 24 June 2026

References

1. Rogalski, B., Dawson, B., Heasman, J. & Gabbett, T. J. Training and game loads and injury risk in elite Australian footballers. *J. Sci. Med. sport*. **16**, 499–503. https://doi.org/10.1016/j.jsams.2012.12.004 (2013).
2. Jaspers, A. et al. Relationships between the external and internal training load in professional soccer: what can we learn from machine learning? *Int. J. Sports Physiol. Perform.* **13**, 625–630. https://doi.org/10.1123/ijspp.2017-0299 (2018).
3. Zech, A. et al. Sex differences in injury rates in team-sport athletes: A systematic review and meta-regression analysis. *J. sport health Sci.* **11**, 104–114. https://doi.org/10.1016/j.jshs.2021.04.003 (2022).
4. Knowles, S. B. et al. Cost of injuries from a prospective cohort study of North Carolina high school athletes. *Injury prevention: J. Int. Soc. Child. Adolesc. Injury Prev.* **13**, 416–421. https://doi.org/10.1136/ip.2006.014720 (2007).
5. Filbay, S. R., Culvenor, A. G., Ackerman, I. N., Russell, T. G. & Crossley, K. M. Quality of life in anterior cruciate ligament-deficient individuals: a systematic review and meta-analysis. *Br. J. Sports Med.* **49**, 1033–1041. https://doi.org/10.1136/bjsports-2015-094864 (2015).
6. Ben Saad, H. et al. Injury prediction in football: how artificial intelligence is shaping the present and transforming the future in Africa. *Sports Health: Multidisciplinary Approach*. **18**, 247–249. https://doi.org/10.1177/19417381251410084 (2026).
7. Bittencourt, N. F. N. et al. Complex systems approach for sports injuries: moving from risk factor identification to injury pattern recognition-narrative review and new concept. *Br. J. Sports Med.* **50**, 1309–1314. https://doi.org/10.1136/bjsports-2015-095850 (2016).
8. Huang, Y. et al. A novel lower extremity non-contact injury risk prediction model based on multimodal fusion and interpretable machine learning. *Front. Physiol.* **13**, 937546. https://doi.org/10.3389/fphys.2022.937546 (2022).
9. Claudino, J. G. et al. Current approaches to the use of artificial intelligence for injury risk assessment and performance prediction in team sports: a systematic review. *Sports Med. - open.* **5**, 28. https://doi.org/10.1186/s40798-019-0202-3 (2019).
10. Van Calster, B., Wynants, L., Timmerman, D., Steyerberg, E. W. & Collins, G. S. Predictive analytics in health care: how can we know it works? *J. Am. Med. Inf. Association: JAMIA*. **26**, 1651–1654. https://doi.org/10.1093/jamia/ocz130 (2019).
11. López-Valenciano, A. et al. A preventive model for muscle injuries: A novel approach based on learning algorithms. *Med. Sci. Sports. Exerc.* **50**, 915–927. https://doi.org/10.1249/mss.0000000000001535 (2018).
12. Rommers, N. et al. A machine learning approach to assess injury risk in elite youth football players. *Med. Sci. Sports. Exerc.* **52**, 1745–1751. https://doi.org/10.1249/mss.0000000000002305 (2020).
13. De Ridder, R., Witvrouw, E., Dolphens, M., Roosen, P. & Van Ginckel, A. Hip strength as an intrinsic risk factor for lateral ankle sprains in youth soccer players: A 3-season prospective study. *Am. J. Sports Med.* **45**, 410–416. https://doi.org/10.1177/0363546516672650 (2017).
14. Ko, J., Rosen, A. B. & Brown, C. N. Functional performance tests identify lateral ankle sprain risk: A prospective pilot study in adolescent soccer players. *Scand. J. Med. Sci. Sports*. **28**, 2611–2616. https://doi.org/10.1111/sms.13279 (2018).
15. Henriquez, M., Sumner, J., Faherty, M., Sell, T. & Bent, B. Machine learning to predict lower extremity musculoskeletal injury risk in student athletes. *Front. sports Act. living*. **2**, 576655. https://doi.org/10.3389/fspor.2020.576655 (2020).
16. Taborri, J. et al. A machine-learning approach to measure the anterior cruciate ligament injury risk in female basketball players. *Sensors*. **21** https://doi.org/10.3390/s21093141 (2021).
17. Bullock, G. S. et al. Just how confident can we be in predicting sports injuries? A systematic review of the methodological conduct and performance of existing musculoskeletal injury prediction models in sport. *Sports Med.* **52**, 2469–2482. https://doi.org/10.1007/s40279-022-01698-9 (2022).
18. Leckey, C., van Dyk, N., Doherty, C., Lawlor, A. & Delahunt, E. Machine learning approaches to injury risk prediction in sport: a scoping review with evidence synthesis. *Br. J. Sports Med.* **59**, 491–500. https://doi.org/10.1136/bjsports-2024-108576 (2025).
19. Dhahbi, W. & Chamari, K. The algorithmic athlete: A call to standardize assessment of sensor technologies and artificial intelligence. *Int. J. Sports Physiol. Perform.* **21**, 505–506. https://doi.org/10.1123/ijspp.2025-0547 (2026).
20. Rayner, D. G., Kim, B. & Foroutan, F. A brief step-by-step guide on conducting a systematic review and meta-analysis of prognostic model studies. *J. Clin. Epidemiol.* **170**, 111360. https://doi.org/10.1016/j.jclinepi.2024.111360 (2024).
21. Page, M. J. et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* **372**, n71. https://doi.org/10.1136/bmj.n71 (2021).
22. Collins, G. et al. Protocol for development of a reporting guideline (TRIPOD-AI) and risk of bias tool (PROBAST-AI) for diagnostic and prognostic prediction model studies based on artificial intelligence. *BMJ Open.* **11**, e048008. https://doi.org/10.1136/bmjopen-2020-048008 (2021).
23. Rutter, C. M. & Gatsonis, C. A. A hierarchical regression approach to meta-analysis of diagnostic test accuracy evaluations. *Stat. Med.* **20**, 2865–2884. https://doi.org/10.1002/sim.942 (2001).
24. Lövdal, S. S., Den Hartigh, R. J. R. & Azzopardi, G. Injury prediction in competitive runners with machine learning. *Int. J. Sports Physiol. Perform.* **16**, 1522–1531. https://doi.org/10.1123/ijspp.2020-0518 (2021).
25. Robles-Palazón, F. J. et al. Predicting injury risk using machine learning in male youth soccer players. *Chaos Solitons Fractals*. **167**, 113079. https://doi.org/10.1016/j.chaos.2022.113079 (2023).
26. Lu, Y. et al. Machine learning for predicting lower extremity muscle strain in national basketball association athletes. *Orthop. J. sports Med.* **10**, 23259671221111742. https://doi.org/10.1177/23259671221111742 (2022).
27. Jauhiainen, S. et al. Predicting ACL injury using machine learning on data from an extensive screening test battery of 880 female elite athletes. *Am. J. Sports Med.* **50**, 2917–2924. https://doi.org/10.1177/03635465221112095 (2022).

28. Castellanos, J. et al. Predicting risk of sport-related concussion in collegiate athletes and military cadets: A machine learning approach using baseline data from the CARE Consortium Study. *Sports Med.* **51**, 567–579. https://doi.org/10.1007/s40279-020-01390-w (2021).
29. Luu, B. C. et al. Machine learning outperforms logistic regression analysis to predict next-season NHL player injury: An analysis of 2322 players from 2007 to 2017. *Orthop. J. sports Med.* **8**, 2325967120953404. https://doi.org/10.1177/2325967120953404 (2020).
30. Oliver, J. L. et al. Using machine learning to improve our understanding of injury risk and prediction in elite male youth football players. *J. Sci. Med. sport.* **23**, 1044–1048. https://doi.org/10.1016/j.jsams.2020.04.021 (2020).
31. Mandorino, M., Figueiredo, A. J., Cima, G. & Tessitore, A. Predictive analytic techniques to identify hidden relationships between training load, fatigue and muscle strains in young soccer players. *Sports* **10** https://doi.org/10.3390/sports10010003 (2021).
32. Shaw, A., Newman, P., Witchalls, J. & Hedger, T. Externally validated machine learning algorithm accurately predicts medial tibial stress syndrome in military trainees: a multicohort study. *BMJ Open. Sport Exerc. Med.* **9**, e001566. https://doi.org/10.1136/bmjsem-2023-001566 (2023).
33. Ayala, F. et al. A preventive model for hamstring injuries in professional soccer: Learning algorithms. *Int. J. Sports Med.* **40**, 344–353. https://doi.org/10.1055/a-0826-1955 (2019).
34. Sun, W. et al. Performance of deep learning algorithms to distinguish high-grade glioma from low-grade glioma: A systematic review and meta-analysis. *iScience* **26**, 106815. https://doi.org/10.1016/j.isci.2023.106815 (2023).
35. Riley, R. D. et al. Calculating the sample size required for developing a clinical prediction model. *BMJ.* **368**, m441. https://doi.org/10.1136/bmj.m441 (2020).
36. Efthimiou, O. et al. Developing clinical prediction models: a step-by-step guide. *BMJ.* **386**, e078276. https://doi.org/10.1136/bmj-2023-078276 (2024).
37. Yadav, S. & Shukla, S. Analysis of k-fold cross-validation over hold-out validation on colossal datasets for quality classification. In *Proceedings of the IEEE 6th International Conference on Advanced Computing (IACC)*, 78–83 (2016).
38. Altman, D. G. & Royston, P. What do we mean by validating a prognostic model? *Stat. Med.* **19**, 453–473. https://doi.org/10.1002/(sici)1097-0258(20000229)19:4%3C453::aid-sim350%3E3.0.co;2-5 (2000).
39. Steyerberg, E. W. et al. Internal validation of predictive models: efficiency of some procedures for logistic regression analysis. *J. Clin. Epidemiol.* **54**, 774–781. https://doi.org/10.1016/s0895-4356(01)00341-9 (2001).
40. Tavanaei, R. et al. Performance of Radiomics-based machine learning and deep learning-based methods in the prediction of tumor grade in meningioma: a systematic review and meta-analysis. *Neurosurg. Rev.* **48** https://doi.org/10.1007/s10143-025-03236-3 (2025).
41. Bleeker, S. E. et al. External validation is necessary in prediction research: a clinical example. *J. Clin. Epidemiol.* **56**, 826–832. https://doi.org/10.1016/s0895-4356(03)00207-5 (2003).
42. Debray, T. P. et al. A new framework to enhance the interpretation of external validation studies of clinical prediction models. *J. Clin. Epidemiol.* **68**, 279–289. https://doi.org/10.1016/j.jclinepi.2014.06.018 (2015).
43. Collins, G. S. et al. External validation of multivariable prediction models: a systematic review of methodological conduct and reporting. *BMC Med. Res. Methodol.* **14** https://doi.org/10.1186/1471-2288-14-40 (2014).
44. Dhiman, P. et al. Reporting of prognostic clinical prediction models based on machine learning methods in oncology needs to be improved. *J. Clin. Epidemiol.* **138**, 60–72. https://doi.org/10.1016/j.jclinepi.2021.06.024 (2021).
45. van der Does, H. T., Brink, M. S., Otter, R. T., Visscher, C. & Lemmink, K. A. Injury risk is increased by changes in perceived recovery of team sport players. *Clin. J. Sport Med.* **27**, 46–51. https://doi.org/10.1097/jsm.0000000000000306 (2017).
46. Dhahbi, W. & Dergaa, I. Machine learning with temporal control designs for testing rhythmic specificity in chronobiology: A multivariate framework proposal for distinguishing genuine biological rhythms from temporal artifacts. *Chronobiol. Int.* https://doi.org/10.1080/07420528.2026.2650826 (2026).
47. Dhahbi, W., Ben Saad, H., Dergaa, I., Souaifi, M. & Chamari, K. Injury profiling in male police cadets during initial training phase: A retrospective cohort study. *Am. J. Men's Health*. **18**, 15579883241304584. https://doi.org/10.1177/15579883241304584 (2024).

Author contributions

S.L.: Writing– review & editing, Writing original draft, Methodology, Formal analysis, Data curation, Conceptualization. G.S.: Writing– review &editing, Methodology, Data curation, Conceptualization. H.Z.: Writing– review & editing, Methodology, Data curation. C.S.: Writing– review & editing, Methodology, Data curation. Z.H. and B.Y.: Writing– review & editing, Methodology, Data curation. All authors reviewed and approved the final version of the manuscript.

Funding

The author(s) declare that financial support was received for the research and/or publication of this article. This work was supported by The National Social Science Fund of China (No. 24CTY025), the Humanities and Social Sciences Youth Foundation, Ministry of Education of the People's Republic of China (No. 22YJC890056) and the "Pioneer" and "Leading Goose" R&D Program of Zhejiang (No. 2024C03260 and No. 2026C01028).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-59571-y.

Correspondence and requests for materials should be addressed to C.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026